Lesson Planning

Complete Lesson Plan Book

Class: Class 7

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Integers: Adding and Subtracting

DM

Unit 1: Integers: Adding and Subtracting

Learning Objectives

• Students will be able to add and subtract integers, including negative numbers.

• Students will be able to use the number line to find the sum or difference of integers.

• Students will be able to apply the sign rules: adding a negative moves left, subtracting a negative moves right.

• Students will be able to solve real-life problems involving positive and negative numbers.

Key Concepts

• Integers are the whole numbers ... -3, -2, -1, 0, 1, 2, 3 ... ; negatives lie left of zero on the number line.

• Adding a positive integer moves right; adding a negative integer moves left: 4 + (-3) = 1.

• Subtracting an integer is the same as adding its opposite: 5 - (-2) = 5 + 2 = 7.

• When signs are the same, add the numbers and keep the sign; when signs differ, subtract and take the sign of the larger one: -6 + 4 = -2.

Teaching Methodology (Step by Step)

1. Open with a 5-minute revision: mark -5 to +5 on a number line drawn on the board.
2. Show addition with a number-line walk: start at 2, add -5 (walk left 5 steps), land on -3.
3. Present the two sign rules with worked examples: (-4) + 6 = 2 and (-4) + (-6) = -10.
4. Turn subtraction into addition of the opposite: 3 - 8 = 3 + (-8) = -5; 3 - (-8) = 3 + 8 = 11.
5. Students solve 6 mixed sums on mini whiteboards while the teacher checks each row.
6. Real-life link: temperature drop from 3°C to -2°C (change = -2 - 3 = -5, a fall of 5 degrees).
7. Close with a 5-question rapid-fire quiz and correct common sign errors on the board.

Classroom Activities

• Number-line walk: students physically step forward and backward to solve integer sums.

• Sign-rule card sort: pairs sort mixed sums into add-same-sign and add-different-sign groups.

• Integer bingo: the teacher calls sums like -7 + 3 and students mark answers on bingo cards.

Assessment Questions

1. Compute: (-9) + 4; 6 + (-11); (-5) + (-8).
2. Compute: 7 - 12; (-3) - (-9); 0 - 6.
3. Show on a number line that 2 + (-7) = -5.
4. The temperature was 4°C at noon and -3°C at night. By how many degrees did it fall?

Homework

• Solve 10 mixed integer addition and subtraction sums.

• Write two real-life situations that involve negative numbers and solve each.

• Draw a number line and show the steps for (-6) - 4 and (-6) - (-4).

Time Allocation

Two periods (80 minutes): 10 min number-line revision, 20 min addition rules with number-line walks, 15 min subtraction-as-addition, 20 min practice and activities, 10 min quiz, 5 min homework briefing.